「Thm. Let $f:[a,b]\to\mathbb{R}$ be a Riemann-integrable, bounded map.
Then, $$F:[a,b]\to\mathbb{R};\ x\mapsto\int_a^x f(t)\,dt$$
is continuous uniformly. Moreover, if f is continuous at $x_0\in[a,b]$, then the derivative
$$F'(x_0)=f(x_0) \text{ exists.}$$

Proof. Let $M>0$ be a bound for $|f|$. Then, given $x,y\in[a,b]$,
$$|F(x)-F(y)|=\left|\int_y^x f(t)\,dt\right|\le\int_y^x|f(t)|\,dt\le M\cdot|x-y|.$$
Thus, given $\varepsilon>0$, we can take $\delta=\frac{\varepsilon}{M}$, to uniformly continuity.

Now, let $x_0\in[a,b]$ s.t. f is continuous at x_0.
Given $\varepsilon>0$, take $\delta>0$ such that
$$|x-x_0|<\delta\Rightarrow|f(x)-f(x_0)|<\varepsilon$$
Now, if $0<|x-x|<\delta$, then
$$\left|\frac{F(x)-F(x_0)}{x-x_0}-f(x_0)\right|=\left|\frac{1}{x-x_0}\cdot\left(\int_{x_0}^x f(t)\,dt\right)-f(x_0)\right|$$
$$=\left|\frac{1}{x-x_0}\cdot\int_{x_0}^x f(t)-f(x_0)\,dt\right|<\varepsilon.$$

「Thm. If $f:[a,b]\to\mathbb{R}$ is Riemann-integrable and $F'=f$, then
$$\int_a^b f(t)\,dt=F(b)-F(a).$$

Proof. Let $P=\{x_0,\cdots,x_n\}$ be an arbitrary partition on $[a,b]$.
Then, by the mean-value Theorem, for each $1\le i\le n$,
$$F(x_i)-F(x_{i-1})=F'(t_i)\cdot(x_i-x_{i-1}) \text{ for some } t_i\in(x_{i-1},x_i).$$
So, $$F(b)-F(a)=\sum_{k=1}^n\left(F(x_i)-F(x_{i-1})\right)=\sum_{k=1}^n f(t_i)\cdot(x_i-x_{i-1}).$$
Now, since $$L(f,P)\le\sum_{k=1}^n f(t_i)\cdot(x_i-x_{i-1})\le U(f,P),$$
the squeeze Theorem gives the result.